TLS Final Training & Operational Manual (Preliminary Version)

1. Introduction & Purpose

TLS provides precision approach guidance where ILS cannot be installed. It creates synthetic ILS signals by processing aircraft transponder replies.

2. Antennas & Units

ESA: Elevation, ASA: Azimuth, ATA: Tracking, LOC: Localizer, GS: Glide Slope, GTU: Generates synthetic ILS signals, CAL/BIT: Validation & self-check.

3. System Integration

Aircraft transponder → ASA/ESA/ATA → GTU → LOC & GS → Aircraft cockpit instruments. CAL/BIT validates, Router links with ATC.

4. Advantages & Limitations

Advantages: flexible, portable, uses existing transponders. Limitations: depends on transponder accuracy, smaller coverage, processing complexity.

5. Operational Scenarios

Crosswind, Mountainous Terrain, Short/Temporary Runways, Low Visibility.

6. Military Applications

Rapid deployment, flexibility in harsh environments, mobility for security, compatibility with mixed fleets, operational advantage in combat zones.

Diagram 1: Overall TLS System Flow

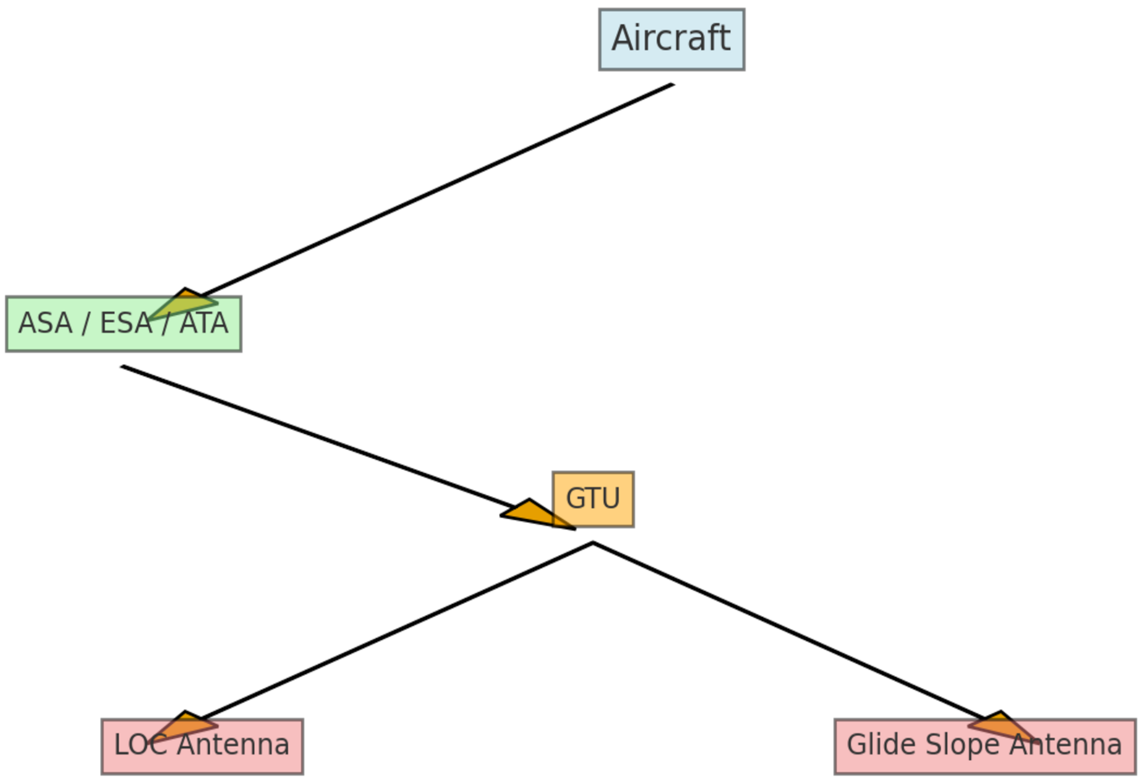


Diagram 2: ESA, ASA, and ATA Roles

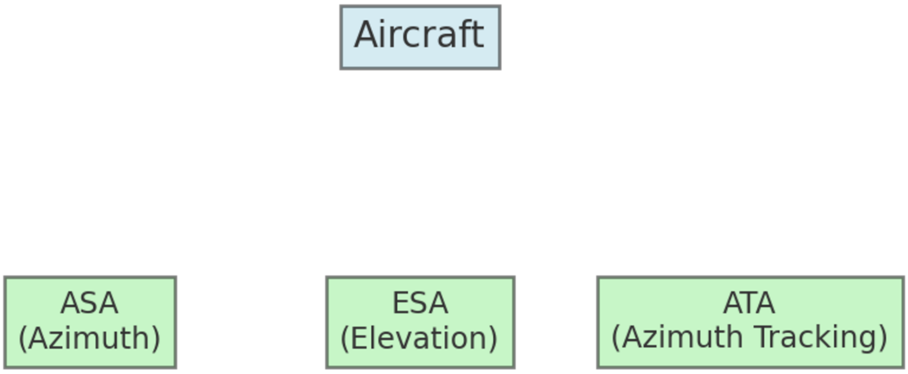


Diagram 3: GTU Processing

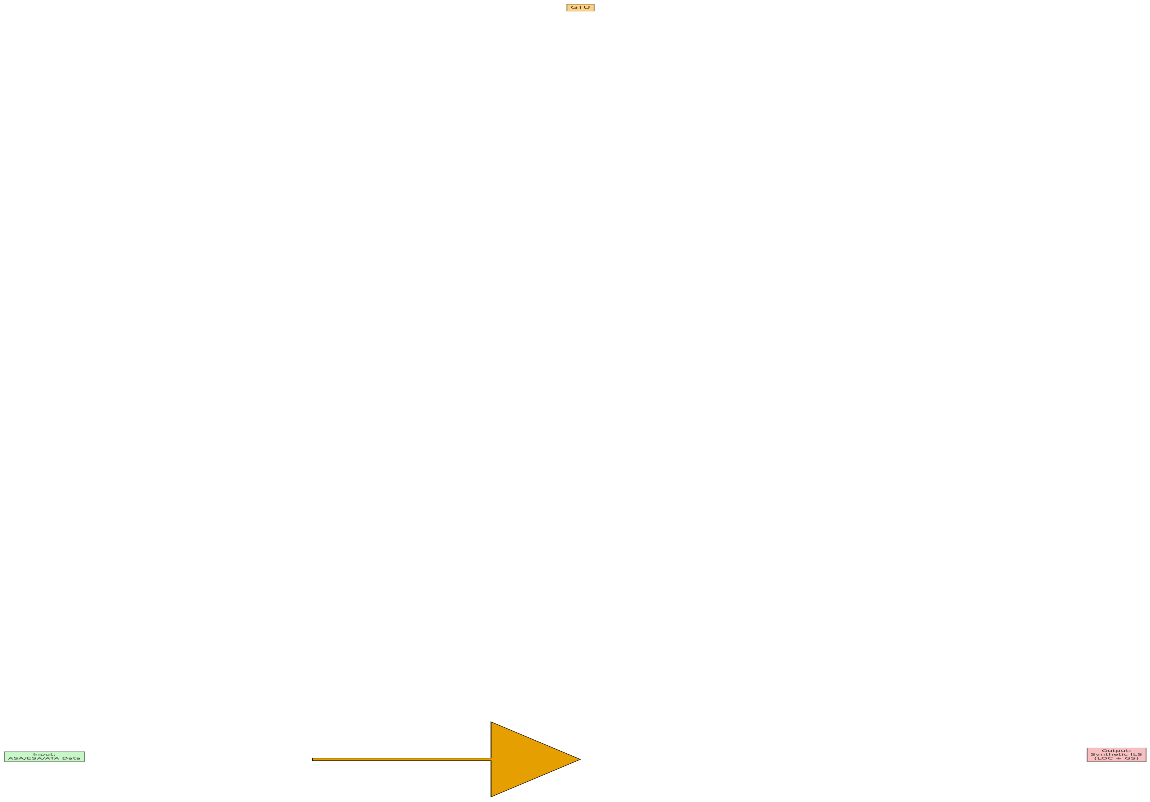


Diagram 4: LOC and Glide Slope Antennas

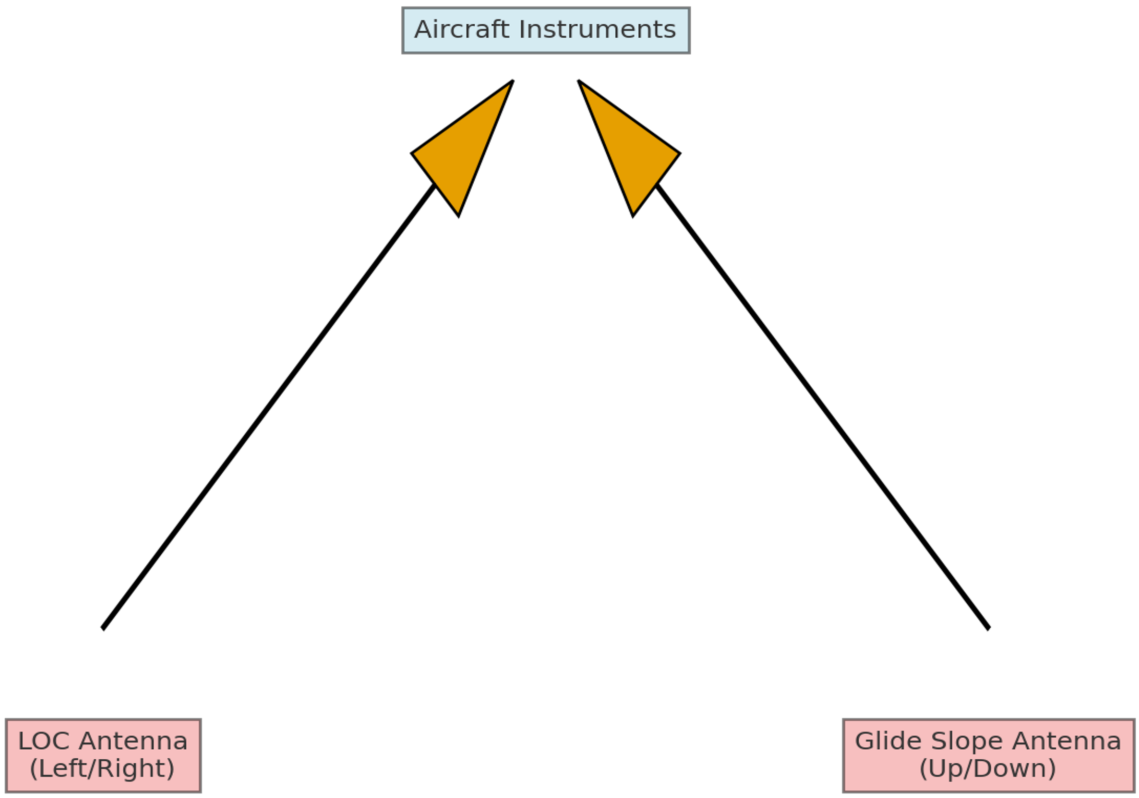


Diagram 5: CAL/BIT Monitoring

Diagram 6: Crosswind Landing Correction

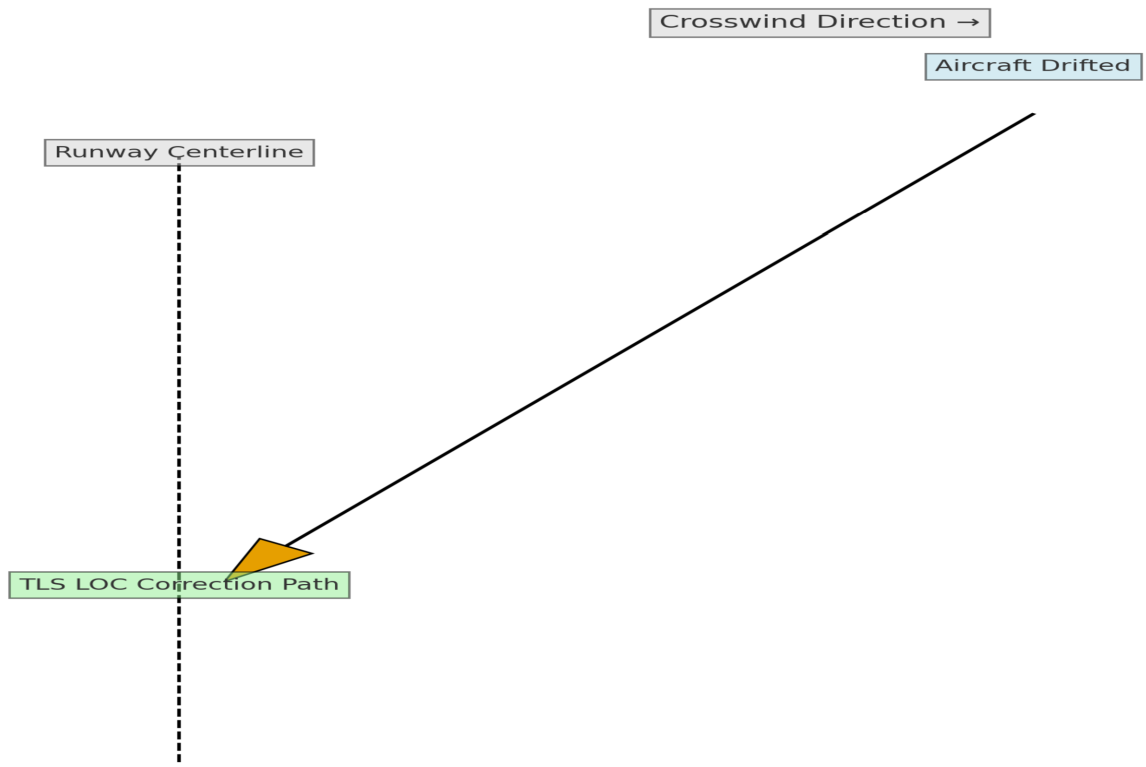


Diagram 7: TLS in Mountainous Terrain

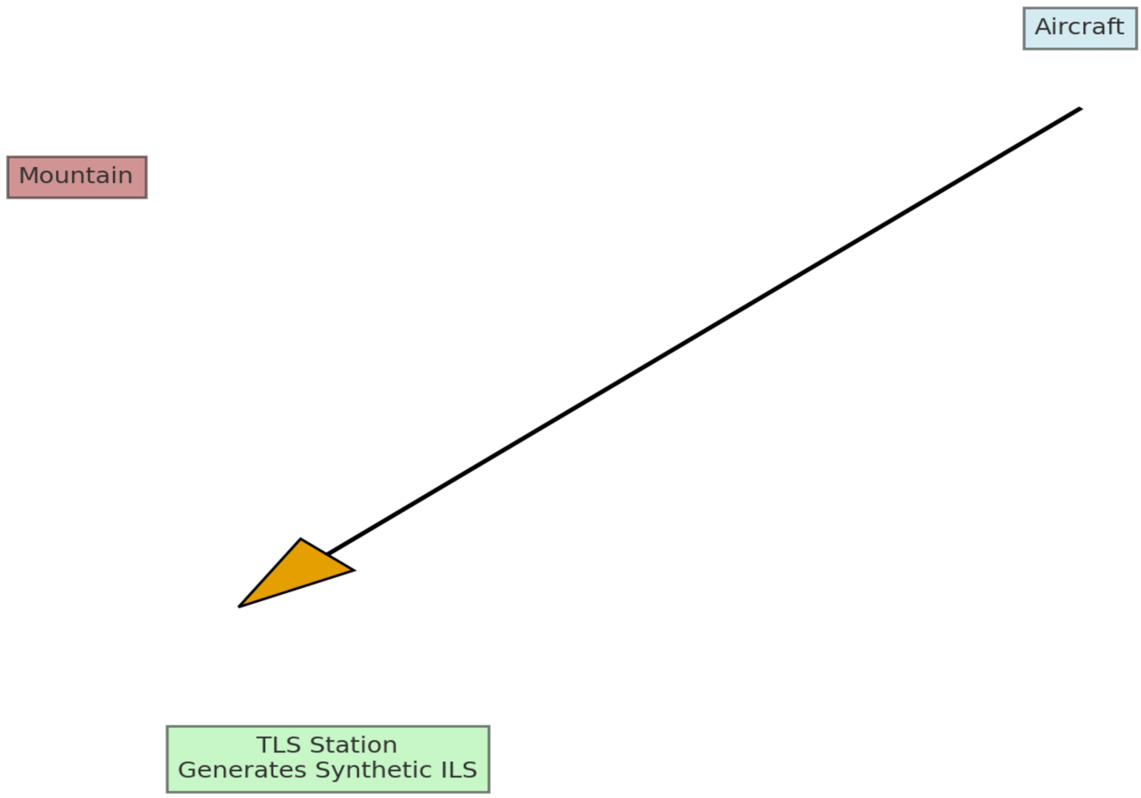


Diagram 8: TLS at Temporary Runway

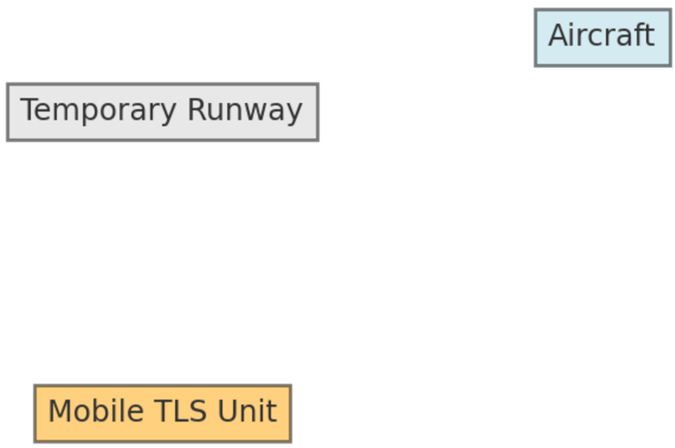


Diagram 9: TLS in Low-Visibility Conditions

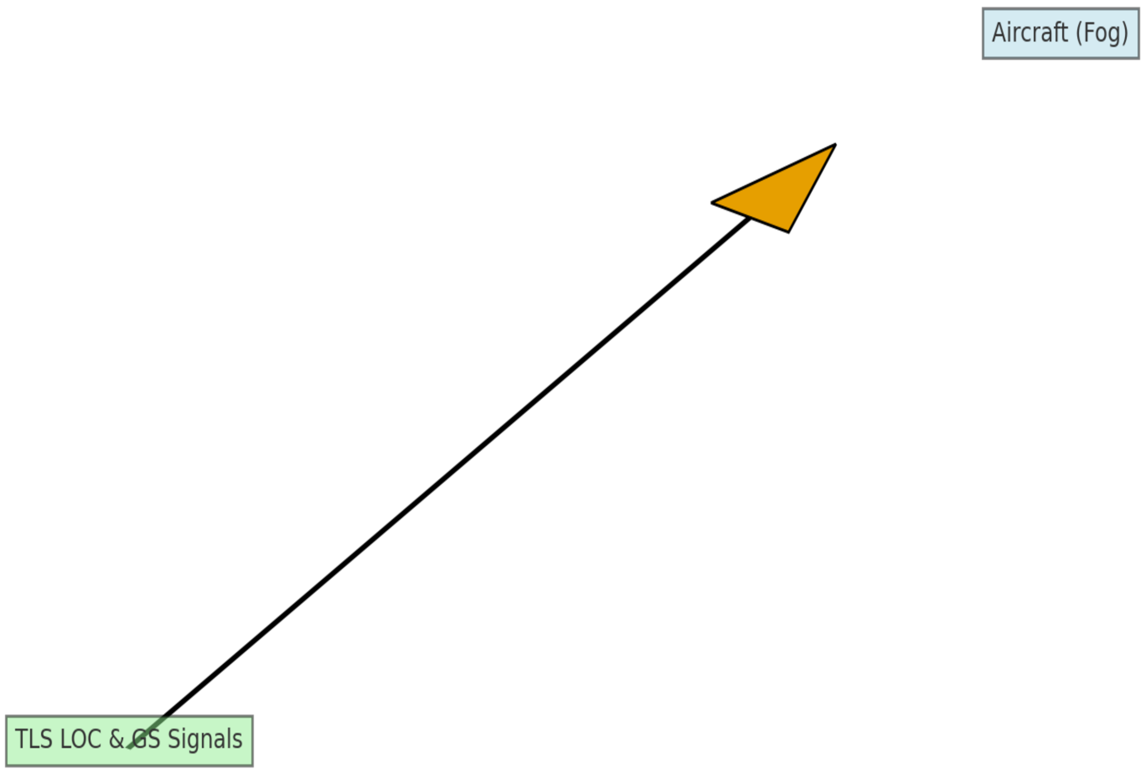
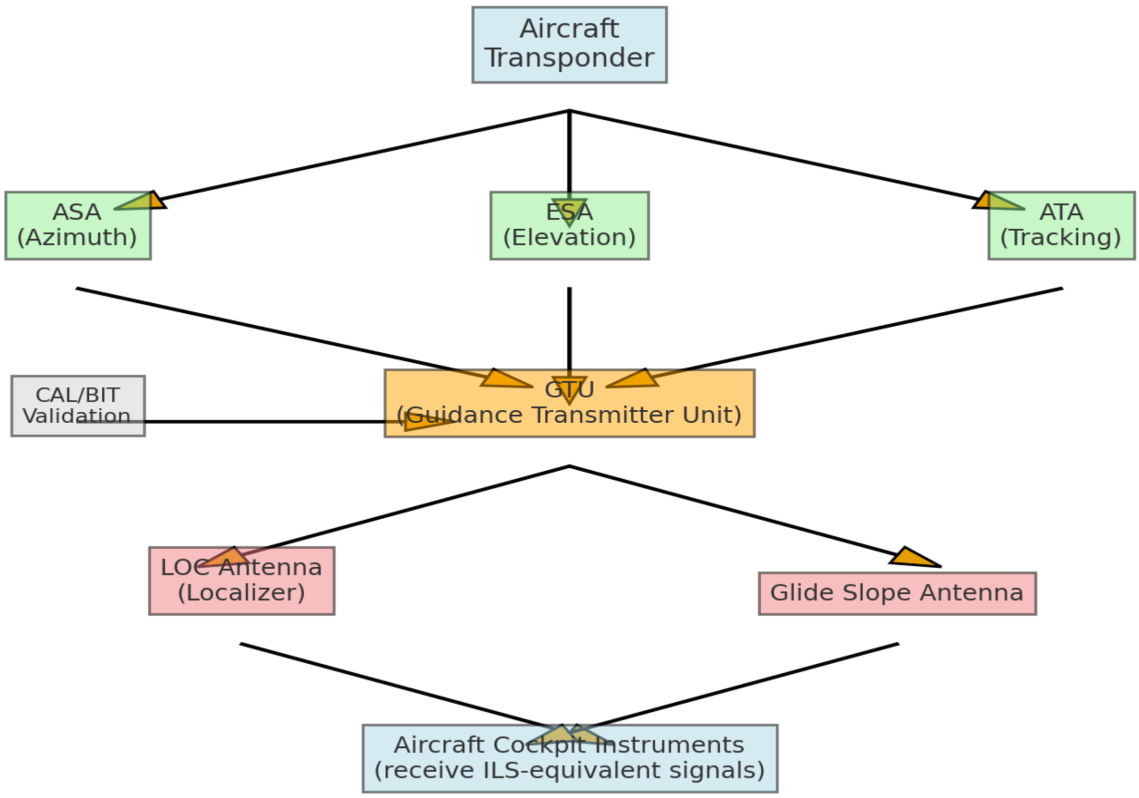


Diagram 10: Full TLS Signal Flow



Appendix: Glossary of Terms

TLS: Transponder Landing System
ILS: Instrument Landing System
ESA: Elevation Sector Antenna
ASA: Azimuth Sector Antenna
ATA: Azimuth Tracking Antenna
LOC: Localizer Antenna
GS: Glide Slope Antenna
GTU: Guidance Transmitter Unit
CAL/BIT: Calibration / Built-in Test
ATC: Air Traffic Control

Review Questions with Answers

Q1: What is the main difference between TLS and ILS?
A1: ILS uses fixed antennas; TLS generates synthetic ILS signals from transponder data.

Q2: Which unit generates ILS-equivalent signals?
A2: GTU (Guidance Transmitter Unit).

Q3: Which antennas help correct aircraft drift in Crosswind?
A3: ASA and ATA.

Q4: What does CAL/BIT mean?
A4: Calibration / Built-in Test, ensures system accuracy.

Q5: Mention two TLS limitations compared to ILS.
A5: Depends on transponder accuracy, smaller coverage, processing complexity.